Eric Nai-Li Chen

Boston, MA eric_n_chen@alumni.brown.edu ericnlchen.github.io ericnlchen

Education

Brown University, MS in Computer Science — Sept 2023 – May 2025
Providence, RI
Advisor: Jeff Huang

UCLA, BS in Computer Science & Minor in Mathematics — Sept 2018 – June 2023
Los Angeles, CA
GPA: 3.84

Research Experience

Boston University, Visiting Researcher — Boston, MA
Mentor: Chang Xiao — Jan 2026 - Present

Adobe, Research Intern — San Jose, CA
Mentors: Li-Yi Wei, Rubaiat Habib Kazi — Jun 2025 – Sep 2025

- Designed the Actionbrushes interaction model for authoring elemental dynamic effects with a generative brush-and-palette interface
- Prototyped proof-of-concept system enabling generation of high-level semantic sliders linked to low-level particle system parameters

Brown HCI Lab, Research Assistant — Providence, RI
Advisor: Jeff Huang — Feb 2024 - May 2025

- Personally invented and implemented L.ink: an L-system-based drawing tool empowering artists to create animated organic structures with a single stroke
- Designed direct-manipulation rule editor and hand-drawn stamp features for intuitive L-system control
- Conducted 12-participant user study investigating how the combination of controllability and surprise impacts creative process and workflow

NASA Academic Mission Services (USRA), Data Science Intern — Mountain View, CA
Mentors: David Bell, Aditya Das, Milad Memarzadeh — Jun 2020 - Sep 2021

- Led multi-organizational project creating wildfire machine learning models in collaboration with Civil Air Patrol captain, NASA research scientist, and USRA RIACS Director, presenting findings in 3 first-author posters at NASA Ames research symposium

Publications

CONFERENCE PAPERS

L.ink: Procedural Ink Growth for Controllable Surprise — Sept 2025
Eric Nai-Li Chen, Joshua Kong Yang, Jeff Huang, Tongyu Zhou
Proc. ACM User Interface Software and Technology (UIST '25)
10.1145/3746059.3747702

EXTENDED ABSTRACTS

Actionbrushes: Painting with Elemental Dynamics from a Generative Palette — April 2025
Eric Nai-Li Chen, Rubaiat Habib Kazi, Li-Yi Wei

Extended Abstracts ACM Human Factors in Computing Systems (CHI EA ’26)
10.1145/3772363.3798553

L.ink: Illustrating Controllable Surprise with L-System Based Strokes April 2025
Eric Nai-Li Chen, Tongyu Zhou, Joshua Kong Yang, Jeff Huang
Extended Abstracts ACM Human Factors in Computing Systems (CHI EA ’25)
10.1145/3706599.3720069

Invited Talks

UCSD Design Lab Seminar Oct 2025
Continuous Creative Channels Between Artist and Medium

Academic Service

2026 CHI (EA Reviewer), C&C (Paper Reviewer), DIS (Paper Reviewer)

2025 CHI (EA Reviewer)

Selected Projects

SplatBrush: XR Painting with 3D Gaussians — CSCI 2951-I: Computer Vision for Graphics and Interaction

- Designed WebXR palette and brush interface for interactive 3D painting with Gaussian textures optimized from real-world materials
- Gathered multi-view material dataset using custom capture ring

Differentiable Rendering of Signed Distance Fields — CSCI 2240: Advanced Computer Graphics

- Implemented the method of Wang et al., *A Simple Approach to Differentiable Rendering of SDFs*
- Produced animated visualizations of gradient descent in action by optimizing scene parameters including lights, materials, object transformations, and freeform geometry

Video Analogies — CSCI 1290: Computational Photography

- Implemented patch-based style transfer from images to videos, extending Hertzmann et al., *Image Analogies*

Selected Coursework

Brown University: Computer Graphics, Advanced Computer Graphics, Computational Photography, Computer Vision for Graphics & Interaction, Deep Learning, Interaction Design

Rhode Island School of Design: Drawing with Computers

UCLA: Software Engineering, Programming Languages, Formal Languages & Automata Theory, Artificial Intelligence, Machine Learning, Database Management Systems, Computer Networks, Operating Systems, Algorithms & Complexity, Real Analysis, Set Theory, Category Theory

Skills

Languages: JavaScript/TypeScript, HTML/CSS, C/C++, Python

Technologies: React, Three.js, OpenGL, TensorFlow, Express, Flask

Other: Public speaking, Leadership, 4th Degree Black Belt and ATA Taekwondo World Champion